

Transformers as Statisticians

Provable In-Context Learning with In-Context Algorithm Selection

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- Motivation and Background

- Transformers have displayed an impressive ability to solve tasks not encountered at pre-training time without explicitly taking a gradient step
 - This phenomenon is dubbed "in-context learning" (Garg et al., 2022)
- Other recent works have shown that Transformers are capable of simulating gradient descent in the context of linear regression (Akyürek et al., 2023; von Oswald et al., 2023)
 - The former requires a lengthy construction of Transformer weights
 - Both only consider a single step of GD
 - This work goes beyond these by proving simulation of GD in the context of GLM, LASSO, and 2-layer NNs as well

- Outline

- Formalism for In-Context Learning with Transformers
- In-Context Simulation of Gradient Descent
- In-Context Algorithm Selection
- Generalization Bounds for Pre-Training

Formalism for In-Context Learning

- The Transformer Model

Def (Attention Layer): A self-attention layer with M heads with parameters $\Theta = \{(V_m, Q_m, K_m)\}_{m \in [M]}$ is defined by

$$\tilde{H} = \text{Attn}_{\Theta}(H) = H + \frac{1}{M} \sum_{m=1}^M (V_m H) \sigma((Q_m H)^T (K_m H)) \in \mathbb{R}^{D \times N},$$

$\uparrow$ $\in \mathbb{R}^{D \times N}$ $\uparrow$ $\in \mathbb{R}^{D \times D}$

$$\tilde{h}_i = [\text{Attn}_{\Theta}(H)]_i = h_i + \sum_{m=1}^M \frac{1}{M} \sum_{j=1}^N \sigma(\langle Q_m h_i, K_m h_j \rangle) V_m h_j$$

$\in \mathbb{R}^{D \times D}$ $\in \mathbb{R}^{D \times D}$

Def (MLP): $\text{MLP}_{\Theta}(H) = H + W_2 \sigma(W_1 H)$

$\uparrow$ "residual connection"

Def (Transformer): $H^{(l)} := \text{MLP}_{\Theta_{\text{mlp}}^{(l)}}(\text{Attn}_{\Theta_{\text{attn}}^{(l)}}(H^{(l-1)}))$, $l \in [L]$.

$\text{TF}_{\Theta}(H^{(0)}) = H^{(L)}$. $\Theta = (\Theta_{\text{attn}}^{(1:L)}, \Theta_{\text{mlp}}^{(1:L)})$.

-The Data

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N \stackrel{\text{iid}}{\sim} P \quad \text{Prompt: } (\mathcal{D}, x_{N+1})$$

$$x_i \in \mathbb{R}^d \quad y_i \in \mathbb{R} \quad D = \oplus(d) \text{ encoding dimension}$$

$$H = \begin{bmatrix} x_1 & x_2 & \dots & x_N & x_{N+1} \\ y_1 & y_2 & \dots & y_N & 0 \\ p_1 & p_2 & \dots & p_N & p_{N+1} \end{bmatrix} \begin{matrix} \} d \\ \} 1 \\ \} D-(d+1) \end{matrix}$$

$$p_i = \begin{bmatrix} 0_{D-(d+3)} \\ 1 \\ \mathbb{1}\{i < N+1\} \end{bmatrix} \leftarrow \text{train/test indicator}$$

$$\text{Notation: } h_i = [x_i; y_i'; 0_{D-(d+3)}; 1; t_i]$$

$$y_i' = y_i, i \in [N]$$

$$t_i = \mathbb{1}\{i < N+1\}$$

$$y_{N+1}' = 0$$

$$TF_\theta: H \in \mathbb{R}^{D \times (N+1)} \mapsto \tilde{H} \in \mathbb{R}^{D \times (N+1)}$$

$$\text{Prediction } \hat{y}_{N+1} = [TF_\theta(H)]_{(d+1) \times (N+1)}$$

- Assumptions

- Bounded inputs and labels: $\|x_i\|_2 \leq B_x$ $|y_i| \leq B_y$

Def 12 (Approximability by Sum of ReLUs): Key technical condition for showing that a Transformer layer can approximate a step of GD.

$g: \mathbb{R}^k \rightarrow \mathbb{R}$ is (ϵ, R, M, C) -approximable by sum of ReLUs if there exists $f: \mathbb{R}^k \rightarrow \mathbb{R}$ defined by

attn heads $\rightarrow$ notice how this is already taking $\sigma(QK^T)V$ form

$$f_{M,C}(z) = \sum_{m=1}^M c_m \sigma(a_m^T [z; 1]) \quad \text{with} \quad \sum_{m=1}^M |c_m| \leq C, \quad \max_{m \in [M]} \|a_m\|_1 \leq 1, \quad a_m \in \mathbb{R}^{k+1}, \quad c_m \in \mathbb{R}$$

s.t. $\sup_{z \in [-R,R]^k} |g(z) - f_{M,C}(z)| \leq \epsilon.$

$\nwarrow$ k -dim hypercube

Why is this a reasonable assumption?

- $\rightarrow$ All smooth functions are approximable by sum of ReLUs
- $\rightarrow$ Given bounds on derivatives, we can bound M

Def A.1 (Sufficiently smooth k -variable function): We say a function $g: \mathbb{R}^k \rightarrow \mathbb{R}$ is (R, C_ϵ) -smooth, if for $s = \lceil (k-1)/2 \rceil + 2$, g is a C^s fcn on $B_\infty^k(R)$, and

↑
hypercube, side length $2R$

$$\sup_{z \in B_\infty^k(R)} \|\nabla^i g(z)\|_\infty = \sup_{z \in B_\infty^k(R)} \max_{j_1, \dots, j_i \in [k]} |\partial_{x_{j_1}} \dots \partial_{x_{j_i}} g(x)| \leq L_i$$

for all $i \in \{0, 1, \dots, s\}$ with $\max_{0 \leq i \leq s} L_i R^i \leq C_\epsilon$.

Prop A.1: Let $\epsilon > 0$, $R \geq 1$, $C_\epsilon > 0$. Then any (R, C_ϵ) smooth function is (ϵ, R, M, C) -approximable by sum of ReLUs with

constant depending only on k

$$M \leq \frac{C(k) C_\epsilon^2 \log\left(1 + \frac{C_\epsilon}{\epsilon}\right)}{\epsilon^2}, \quad C \leq C(k) C_\epsilon.$$

In-Context Simulation of Gradient Descent

Prop C.2 (Approximating a Single GD Step by a Single Attention Layer):

let $l(\cdot, \cdot): \mathbb{R}^2 \rightarrow \mathbb{R}$ be a loss function s.t. $\mathcal{A}l$ is (ϵ, R, M, C) -approximable by a sum of ReLUs with $R = \max\{B_x B_w, B_y, 1\}$. let $\hat{L}_N(w) := \frac{1}{N} \sum_{i=1}^N l(w^\top x_i, y_i)$ be the empirical risk with the loss l on $\{(x_i, y_i)\}_{i=1}^N$.

Then, $\forall \epsilon > 0 \exists \Theta = \{(Q_m, K_m, V_m)\}_{m \in [M]}$ s.t. for any input sequence taking the form $h_i = [x_i; y_i'; w; 0_{D-2d+3}; 1; t_i]$ with $\|w\|_2 \leq B_w$, it gives output $\tilde{h}_i = [\text{Attn}_\Theta(H)]_i = [x_i; y_i'; \tilde{w}; 0_{D-2d+3}; 1; t_i] \quad \forall i \in [N+1]$ where

$$\|\tilde{w} - (w - \eta \nabla \hat{L}_N(w))\|_2 \leq \epsilon (\eta B_x).$$

Pf: By (ϵ, R, M, C) -approximability, $\exists f: [-R, R]$ s.t.

$$f(s, t) = \sum_{m=1}^M c_m \sigma(a_m s + b_m t + d_m) \quad \text{with} \quad \sum_{m=1}^M |c_m| \leq C, \quad |a_m| + |b_m| + |d_m| \leq 1 \quad \forall m \in [M]$$

$$\text{s.t.} \sup_{(s, t) \in [-R, R]^2} |f(s, t) - \mathcal{A}l(s, t)| \leq \epsilon.$$

For convenience, omit $\mathcal{O}_{D-2d+3}$.

$$h_i = [x_i, y_i, w, 1, t_i] \quad t_i := \mathbb{1}[i < N+1]$$

$$R = \max\{B_x B_w, B_y, 1\} \quad \|x\| \leq B_x \quad \|w\| \leq B_w \quad |y| \leq B_y$$

$$Q_m h_i = \begin{bmatrix} a_m w \\ b_m \\ d_m \\ -2 \\ 0 \end{bmatrix} \quad K_m h_j = \begin{bmatrix} x_j \\ y_j \\ 1 \\ R(1-t_j) \\ 0 \end{bmatrix} \quad V_m h_j = -\frac{(N+1)\gamma c_m}{N} \begin{bmatrix} \mathcal{O}_d \\ 0 \\ x_j \\ \mathcal{O}_2 \end{bmatrix}$$

$$\forall i, j \in [N+1]$$

The answer to the question

↙ "what's so special about Transformers?"

$$\sigma(\langle Q_m h_i, K_m h_j \rangle) = \sigma(a_m w^T x_j + b_m y_j + d_m - 2R(1-t_j))$$

$$= \sigma(a_m w^T x_j + b_m y_j + d_m) \mathbb{1}[t_j = 1]$$

$$\begin{aligned} \text{since } |a_m w^T x_j + b_m y_j + d_m| &\leq |a_m| \|w\| \|x_j\| + |b_m| |y_j| + |d_m| \\ &\leq (|a_m| + |b_m| + |d_m|) R \leq R \end{aligned}$$

$$\Rightarrow \sum_{m=1}^M \sigma(\langle Q_m h_i, K_m h_j \rangle) V_m h_j$$

$$= \left(\sum_{m=1}^M c_m \sigma(a_m w^T x_j + b_m y_j + d_m) \right) \left(-\frac{(N+1)\eta}{N} \mathbb{1}[t_j=1] \begin{bmatrix} 0_{d+1} \\ x_j \\ 0_2 \end{bmatrix} \right)$$

$$= f(w^T x_j, y_j) \left(-\frac{(N+1)\eta}{N} \mathbb{1}[t_j=1] \begin{bmatrix} 0_{d+1} \\ x_j \\ 0_2 \end{bmatrix} \right)$$

$$\tilde{h}_i = [A \hat{H}_{\theta}(H)]_i = h_i + \frac{1}{N+1} \sum_{j=1}^{N+1} \sum_{m=1}^M \sigma(\langle Q_m h_i, K_m h_j \rangle) V_m h_j$$

$$= h_i - \frac{\eta}{N} \sum_{j=1}^N f(w^T x_j, y_j) [0_{d+1}; x_j; 0_2]$$

$$= [x_i; y_i; w; 1; t_i] - \frac{\eta}{N} \sum_{j=1}^N \partial_1 \ell(w^T x_j, y_j) [0_{d+1}; x_j; 0_2] + [0_{d+1}; \vec{\epsilon}; 0_2]$$

$$= [x_i; y_i; w_{\eta}^+ + \vec{\epsilon}; 1; t_i]$$

$$w_{\eta}^+ = w - \eta \nabla \hat{L}_w(w)$$

The error vector $\tilde{\epsilon}$ satisfies

$$\|\tilde{\epsilon}\|_2 = \left\| -\frac{\eta}{N} \sum_{j=1}^N \left(f(\omega^\top x_j, y_j) - \partial_{\epsilon} l(\omega^\top x_j, y_j) \right) x_j \right\|_2$$

$$\leq \frac{\eta}{N} \sum_{j=1}^N \left| f(\omega^\top x_j, y_j) - \partial_{\epsilon} l(\omega^\top x_j, y_j) \right| \|x_j\|_2$$

$$\leq \frac{\eta}{N} N_{\epsilon} B_x = \epsilon \eta B_x$$

What happens to approximation error after multiple GD steps?

lem 14 (Composition of Error for Approximating Convex GD): Suppose $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is convex, L_f -smooth on $\mathbb{B}_2(R)$. Let $\hat{w}^0 = w_{GD}^0 = 0$ and

$$\begin{cases} \hat{w}^{L+1} = \hat{w}^L - \eta \nabla f(\hat{w}^L) + \varepsilon^L, & \|\varepsilon^L\|_2 \leq \varepsilon \\ w_{GD}^{L+1} = w_{GD}^L - \eta \nabla f(w_{GD}^L) \end{cases}$$

for all $L \geq 0$. Then as long as $\eta \leq 2/L_f$, for any $0 \leq L \leq R/(2\varepsilon)$, it holds that $\|\hat{w}^L - w_{GD}^L\|_2 \leq L\varepsilon$.

Key takeaway: Error grows linearly in the number of steps.

Pf: let $T_\eta: w \mapsto w - \eta \nabla f(w)$

$\forall w, w' \in \mathbb{B}_2(R)$, we have

$$\|T_\eta(w) - T_\eta(w')\|_2^2 = \|w - \eta \nabla f(w) - (w' - \eta \nabla f(w'))\|_2^2$$

$$= \|w - w'\|_2^2 - 2\eta \langle w - w', \nabla f(w) - \nabla f(w') \rangle + \eta^2 \|\nabla f(w) - \nabla f(w')\|_2^2$$

since $\langle w - w', \nabla f(w) - \nabla f(w') \rangle \geq \frac{1}{L_f} \|\nabla f(w) - \nabla f(w')\|_2^2$ for L -smooth convex fns

$$\leq \|w - w'\|_2^2 - (2\eta/L_f - \eta^2) \|\nabla f(w) - \nabla f(w')\|_2^2$$

$$\leq \|w - w'\|_2^2$$

$\Rightarrow T_\eta$ is nonexpansive.

The claim is clear for $L=0$. Suppose it holds for some $L \geq 0$. For $L+1 \leq R/(2\varepsilon)$,

$$\|\hat{w}^{L+1} - w_{GD}^{L+1}\|_2 = \|T_\eta(\hat{w}^L) + \varepsilon^L - T_\eta(w_{GD}^L)\|_2$$

$$\leq \|T_\eta(\hat{w}^L) - T_\eta(w_{GD}^L)\|_2 + \|\varepsilon^L\|_2$$

$$\leq \|\hat{w}^L - w_{GD}^L\|_2 + \varepsilon$$

$$\leq L\varepsilon + \varepsilon = (L+1)\varepsilon.$$

□

Let's combine the previous two results.

Thm 13 (Convex ICGD): Fix any $B_w > 0$, $L > 1$, $\eta > 0$, $\varepsilon \leq B_w/2L$. Assume

(1) The loss $\ell(\cdot, \cdot)$ is convex in the first argument.

(2) ℓ is (ε, R, M, C) -approximable by sum of ReLUs with $R = \max\{B_x B_w, B_y, 1\}$.

Then there exists an attention-only Transformer TF_θ^D with $(L+1)$ layers, $\max_{l \in [L]} M^{(l)} \leq M$ heads and $M^{(L+1)} = 2$ s.t. for any input data $(\mathcal{D}, x_{N+1})$ with

L -smoothness needed for lemma 14

existence of a bd solution

$$\sup_{\|w\|_2 \leq B_w} \lambda_{\max}(\nabla^2 \hat{L}_w(w)) \leq \frac{2}{\eta}, \quad \exists w^* \in \arg \min_{w \in \mathbb{R}^d} \hat{L}_w(w) \text{ s.t. } \|w^*\|_2 \leq B_w/2,$$

(a) (Parameter space) $\|\hat{w}^L - w_{GD}^L\|_2 \leq \varepsilon (L\eta B_x) \quad \forall L \geq 1.$

(b) (Prediction space) $|\hat{y}_{N+1} - \langle w_{GD}^L, x_{N+1} \rangle| \leq \varepsilon (L\eta B_x^2).$

Pf of (a): By previous prop, $\exists$ attention layer with $\theta^{(1)} = \{(Q_m, K_m, V_m)\}_{m \in \mathcal{M}}$ which maps w to $\hat{w}$ s.t.

$$\|\hat{w} - (w - \eta \nabla \hat{L}_w(w))\|_2 \leq \varepsilon \cdot (\eta B_x) =: \varepsilon'.$$

Construct L -layer Transformer $\theta^{1:L} = (\theta^{(1)}, \dots, \theta^{(L)})$.

Use induction and Lemma 14 to show $\|\hat{w}^L - w_{GD}^L\|_2 \leq L\varepsilon' \quad \forall L \in [L].$

□

Pf of (b): Done by constructing the last attention layer $\theta^{(L+1)}$.

Define $\theta^{(L+1)} = \{(Q_m^{(L+1)}, K_m^{(L+1)}, V_m^{(L+1)})\}_{m=1,2}$ s.t. $\forall i, j \in [N+1]$,

$$Q_1^{(L+1)} h_i^{(L)} = [x_i; 0_{D-d}], \quad K_1^{(L+1)} h_j^{(L)} = [\hat{w}^{(L)}; 0_{D-d}], \quad V_1^{(L+1)} h_j^{(L)} = [0_d; 1; 0_{D-d-1}]$$

$$Q_2^{(L+1)} h_i^{(L)} = [x_i; 0_{D-d}], \quad K_2^{(L+1)} h_j^{(L)} = [-\hat{w}^{(L)}; 0_{D-d}], \quad V_2^{(L+1)} h_j^{(L)} = [0_d; -1; 0_{D-d-1}].$$

...

So essentially, the proof that the Transformer can approximate GD in linear regression boils down to selecting the right input embedding and the right parameters for the Transformer layers, which arises naturally from the key assumption that the gradient is approximable by a sum of ReLUs.

There is an additional result (Corollary C.1) which captures the same idea for convex objectives with an additional ℓ_2 regularization term.

Thm 4 (Implementing In-Context Ridge Regression): For any $\lambda > 0$, $0 \leq \alpha \leq \beta$ with $\kappa := \frac{\beta + \lambda}{\alpha + \lambda}$, $B_w > 0$, and $\epsilon < B_x B_w / 2$, there exists an L -layer attention-only Transformer TF_θ^0 with

$$L = \lceil 2\kappa \log(B_x B_w / 2\epsilon) \rceil + 1, \quad \max_{l \in [L]} M^{(l)} \leq 3 \quad \text{s.t. the following holds.}$$

On any input data $(\mathcal{D}, x_{w+1})$ s.t. ridge regression is well-conditioned and has a bounded solution:

$$\alpha \leq \lambda_{\min}(X^T X / N) \leq \lambda_{\max}(X^T X / N) \leq \beta \quad \|w_{\text{ridge}}^\lambda\| \leq B_w / 2,$$

we have

$$|\hat{y}_{w+1} - \langle w_{\text{ridge}}^\lambda, x_{w+1} \rangle| \leq \epsilon \quad \text{and} \quad \|\hat{w}^{L-1} - w_{\text{ridge}}^\lambda\|_2 \leq \epsilon / B_x.$$

We get an expression for the number of layers needed to achieve ϵ error!
This follows from Thm 13 and standard GD convergence results for convex and L -smooth functions.

Pf of Thm 4: let $L_{\text{ridge}}(w) = \frac{1}{2N} \sum_{i=1}^N (\langle w, x_i \rangle - y_i)^2 + \frac{\lambda}{2} \|w\|_2^2$.

We have $\nabla^2 L_{\text{ridge}}(w) = X^T X / N + \lambda I_d$, so L_{ridge} is $(\alpha + \lambda)$ -strongly convex and $(\beta + \lambda)$ -smooth over $\mathbb{R}^d$.

$$w_{\text{GD}}^{t+1} := w_{\text{GD}}^t - \eta \nabla L_{\text{ridge}}(w_{\text{GD}}^t)$$

$$w_{\text{GD}}^0 = 0_d, \quad \eta = \frac{1}{\beta + \lambda}, \quad T := \left\lceil 2\kappa \log\left(\frac{B_x B_w}{2\varepsilon}\right) \right\rceil.$$

By standard GD convergence results for convex and smooth, $\forall t \geq 1$

$$\|w_{\text{GD}}^t - w_{\text{ridge}}^{\lambda}\|_2^2 \leq \exp(-t/\kappa) \|w_{\text{GD}}^0 - w_{\text{ridge}}^{\lambda}\|_2^2 = \exp(-t/\kappa) \|w_{\text{ridge}}^{\lambda}\|_2^2.$$

$$\Rightarrow \|w_{\text{GD}}^T - w_{\text{ridge}}^{\lambda}\|_2^2 \leq \exp(-T/2\kappa) \|w_{\text{ridge}}^{\lambda}\|_2^2 \leq \frac{2\varepsilon}{B_x B_w} \frac{B_w}{2} \leq \frac{\varepsilon}{B_x}$$

Construct a Transformer to approximate w_{GD}^T .

$$l(s, t) := \frac{1}{2} (s - t)^2 \quad \partial_s l(s, t) = s - t$$

$$\partial_s \ell(s, t) = \sigma((s-t)/2) - \sigma(-(s-t)/2) \quad \text{Sum of ReLUs}$$

$\Rightarrow \partial_s \ell(s, t)$ is $(0, R, 2, 4)$ -approximable by a sum of ReLUs $\forall R > 0$.

Apply Corollary C.1 for TF with $L := T+1$ layers s.t.

$$|\hat{y}_{N+1} - \langle W_{GD}^T, x_{N+1} \rangle| = 0.$$

$$\Rightarrow |\hat{y}_{N+1} - \langle W_{ridge}^{\lambda}, x_{N+1} \rangle| = |\langle W_{GD}^T - W_{ridge}^{\lambda}, x_{N+1} \rangle| \leq \frac{\varepsilon}{B_x} B_x = \varepsilon.$$

Cor 5: For any $N \geq \tilde{O}(d)$, $\exists$ an $O(k \log(kN/\sigma))$ -layer Transformer θ s.t. on any P satisfying standard statistical assumptions for least-squares, its ICL prediction $\hat{y}_{N+1}$ achieves

$$\mathbb{E}_{(\mathcal{D}, x_{N+1}, y_{N+1}) \sim P} [(\hat{y}_{N+1} - y_{N+1})^2] \leq \inf_{w \text{ } (x, y) \sim P} \mathbb{E} [(y - \langle w, x \rangle)^2] + \tilde{O}(d\sigma^2/N).$$

Risk of our predictor Risk of best linear predictor

What are the "standard statistical assumptions"?

- $\|x\|_2 \leq B_x$, $|y| \leq B_y$ P -a.s.
- k is the condition number i.e. given $\Sigma_P = \mathbb{E}_P[xx^T]$, $k = \lambda_{\max}(\Sigma_P) / \lambda_{\min}(\Sigma_P)$.
- $\Sigma_P^{-1/2} x$ is K^2 -subgaussian for some $K \geq 1$.
- The best linear predictor w_P^* satisfies $\|w_P^*\|_2 \leq B_w$.
- $\mathbb{E}[(y - \langle x, w_P^* \rangle)^2 | x] \leq \sigma^2$ a.s. (over x).
expected squared residual

Cor 6 (Nearly-Bayes Linear Regression): let $w_* \sim \pi = N(0, I_d/d)$, $x \sim N(0, I_d)$, $y = \langle w_*, x \rangle + N(0, \sigma^2)$. let the Bayes risk

$$R_\pi := \mathbb{E}_{w_* \sim \pi} \mathbb{E}_{\mathcal{D}, x_{N+1}, y_{N+1}} \left[\frac{1}{2} (\hat{y}_{N+1}^{\text{Bayes}} - y_{N+1})^2 \right],$$

where $\hat{y}_{N+1}^{\text{Bayes}}$ is the prediction of the optimal ridge regressor with $\lambda = d\sigma^2/N$.

Then, if $N \geq \max\{d/10, O(\log(1/\epsilon))\}$, $\exists$ an $L = O(\log(1/\epsilon))$ Transformer with

$$\mathbb{E}_{w_* \sim \pi} \mathbb{E}_{\mathcal{D}, x_{N+1}, y_{N+1}} \left[\frac{1}{2} (\hat{y}_{N+1} - y_{N+1})^2 \right] \leq R_\pi + \epsilon.$$

For a sufficiently large sample size, we are nearly Bayes optimal!

The authors then go on to prove very similar results for GLM and LASSO.

- Gradient Descent on Two-Layer Neural Networks (may want to skip)

Two layer NN with scalar pred: $\text{pred}(x; w) = \sum_{k=1}^K \overset{\text{scalar}}{u_k} \overset{\text{vector}}{r(V_k^T x)}$

$$w = [V_k; u_k]_{k \in [K]} \in \mathbb{R}^{K(d+1)}$$

$$\text{ERM: } \min_{w \in W} \hat{L}_N(w) = \frac{1}{2N} \sum_{i=1}^N \ell(\text{pred}(x_i; w), y_i)$$

Assumption: Proj_W can be exactly implemented by an MLP layer.

Thm 6.1 (Approximate ICGD by Two-Layer NNs): Fix any $B_V, B_u > 0$, $L \geq 1$, $\eta > 0$ and $\epsilon > 0$. Suppose that

1) Both the activation function r and the loss function ℓ is C^4 -smooth.

2) W is a closed domain s.t. $W \subseteq \{w: [V_k, u_k]_{k \in [K]} \in \mathbb{R}^{K(d+1)}: \|V_k\| \leq B_V, |u_k| \leq B_u\}$
and $\text{Proj}_W = \text{MLP}_{\Theta_{\text{MLP}}}$ for some MLP layer Θ_{MLP} with hidden dimension Δ_w and $\|\Theta_{\text{MLP}}\| \leq C_w$.

Then there exists a $(2L)$ -layer Transformer TF_0 with

$$\max_{l \in [2L]} M^{(l)} \leq \tilde{O}(\varepsilon^{-2}), \quad \max_{l \in [2L]} D^{(l)} \leq \tilde{O}(\varepsilon^{-2}) + D_w, \quad \|\theta\| \leq O(1+\eta) + C_w,$$

where $\tilde{O}(\cdot)$ hides the constants that depend on k , the radius parameters B_x, B_y, B_u, B_v and the smoothness of r and l , such that for any input data $(\mathcal{D}, x_{N+1})$, $TF_0(H^{(0)})$ approximately implements GD: For every $l \in [L]$, the $2l$ th layer's output $h_i^{(2l)} = [x_i; y_i'; \hat{w}^l; 0; 1; t_i]$, for every $i \in [N+1]$, and

$$\hat{w}^{(l)} = \text{Proj}_w (\hat{w}^{l-1} - \eta (\nabla \hat{L}_w(\hat{w}^{l-1}) + \varepsilon^{l-1})) \quad \hat{w}^0 = 0,$$

where $\|\varepsilon^{l-1}\|_2 \leq \varepsilon$ is an error term.

That is, there exists a Transformer that iteratively performs projected GD.

Cor G.1: For any $L \geq 1$, under the same setting as the thm above, there exists a $(2L)$ -layer Transformer TF_θ s.t.

$$\|\hat{w}^L - w_{\text{GO}}^L\|_2 \leq L_f^{-1} (1 + \eta L_f)^L \varepsilon, \quad \text{exponential in } L \text{ due to non-convexity}$$

$$\text{where } L_f = \sup_{w \in \mathcal{W}} \|\nabla^2 \hat{L}_w(w)\|_{\text{op}}.$$

To prove Thm G.1, it suffices to prove the following:

Prop G.1: Under the assumptions of Thm G.1, there exists a 2-layer Transformer TF_θ with the same bounds on the number of heads, hidden dimension and the norm, such that for any input data $(\mathcal{D}, x_{w+1})$ and any $w \in \mathbb{R}^d$, TF_θ maps

$$h_i = [x_i; y_i; w; 0; |; t_i] \rightarrow h_i' = [x_i; y_i; w_\eta^+; 0; |; t_i], \text{ where}$$

$$w_\eta^+ = \text{Proj}_w(w - \eta \nabla \hat{L}_w(w) + \varepsilon(w)), \quad \|\varepsilon(w)\|_2 \leq \eta \varepsilon.$$

The proof idea is the same as for Thm 13. We construct the Transformer layers to approximate the gradient via Chain Rule:

$$\nabla \hat{L}_N(w) = \frac{1}{N} \sum_{i=1}^N \mathcal{D}_i l(\text{pred}(x_i; w), y_i) \nabla_w \text{pred}(x_i; w)$$

- Use an attention layer to approximate $(x_i, w) \mapsto \text{pred}(x_i; w)$.
- Use MLP to approximate $\text{pred}(x_i; w) \mapsto \mathbb{1}_{\{i \leq N+1\}} \mathcal{D}_i l(\text{pred}(x_i; w), y_i)$
- Use attention layer to compute $w \mapsto w - \eta \nabla \hat{L}_N(w)$
- Use MLP to implement $\text{Proj}_w(\cdot)$

In-Context Algorithm Selection

Post-ICL Validation: Given a prompt containing data, the Transformer can select the model (from a list of K models) that achieves the best validation performance.

Prop 15 (Post-ICL Validation Mechanism): Suppose that ℓ is $(\gamma/3, R, M, C)$ -approx by sum of ReLUs. Then $\exists$ a 3-layer Transformer that maps

$$h_i = \left[\overset{\text{validation data}}{\downarrow} x_i; \overset{\downarrow}{y_i}; *; \overset{\text{fitted "candidate" models}}{\downarrow} f_1(x_i); \dots; f_K(x_i); 0_{K+1}; 1; t_i \right]$$

$$\mapsto \tilde{h}_i = [x_i; y_i; *; \hat{f}(x_i); 1; t_i], \quad i \in [N+1]$$

where $\hat{f}$ is a convex combination of $\{\hat{f}_k : \hat{L}_{\text{val}}(f_k) \leq \min_{k^* \in [K]} \hat{L}_{\text{val}}(f_{k^*}) + \gamma\}$.

$$\hat{L}_{\text{val}}(f) := \frac{1}{|D_{\text{val}}|} \sum_{(x_i, y_i) \in D_{\text{val}}} \ell(f(x_i), y_i),$$

$$\max_{\ell \in [3]} M^{(\ell)} \leq (M+3)K, \quad \max_{\ell \in [3]} D^{(\ell)} \leq K^2 + K + 1.$$

Idea: Construct a layer to evaluate the validation loss of each model.

Then, construct layers which select the best model.

→ Do this by constructing an MLP which takes pairwise differences between losses, i.e. $(L_{f_1} - L_{f_2}, \dots, L_{f_1} - L_{f_k}, \dots, L_{f_k} - L_{f_{k-1}})$.

Example: $f_1, \dots, f_k$ are ridge regression models corresponding to regularization coefficients $\lambda_1, \dots, \lambda_k$.

Thm 16: For $\epsilon > 0$, $\exists$ Transformer with $L = O(\log(1/\epsilon))$ layers and $O(k)$ heads s.t. given $(\mathcal{D}, x_{w+1})$ it outputs $\hat{y}_{w+1} = \langle \hat{w}, x_{w+1} \rangle$ where

$$\text{dist}\left(\hat{w}, \text{conv}\left\{\hat{w}_{\text{ridge}, \text{train}}^{\lambda_k} : \hat{L}_{\text{val}}(\hat{w}_{\text{ridge}, \text{train}}^{\lambda_k}) \leq \min_{k \in [k]} \hat{L}_{\text{val}}(\hat{w}_{\text{ridge}, \text{train}}^{\lambda_k}) + \gamma'\right\}\right) \leq \epsilon,$$

$$\text{with } \gamma' = 2(B_x B_w + B_y) B_x \epsilon + \gamma.$$

Pre-ICH Testing: "We construct a Transformer that performs logistic regression when labels are binary, and linear regression whp if the label admits a continuous distribution."

- Sample Complexity of Pre-Training

In-context learning loss: $Z = (H, y_{N+1})$

$$l_{icl}(\theta; Z) = \frac{1}{2} \left(y_{N+1} - \underbrace{\text{clip}_{B_y}}_{\text{Proj onto } [-B_y, B_y]}(\text{read}_y(TF_{\theta}^R(H))) \right)^2$$

$$TF_{\theta}^R \text{ has } H^{(l)} = \text{clip}_R \left(\text{MLP}_{\theta_{mlp}^{(l)}} \left(\text{Attn}_{\theta_{attn}^{(l)}} (H^{(l-1)}) \right) \right) \quad \forall l \in [L],$$

$$\text{where } \text{clip}_R(H) := \left[\text{Proj}_{\|h\|_2 \leq R} (h_i) \right]_i.$$

used to control Lipschitz const of TF_{θ} wrt θ

Draw $P \sim \pi$ and sample $(x_i, y_i)_{i=1}^{N+1} \stackrel{\text{iid}}{\sim} P^{\otimes N+1}$

$$\hat{L}_{icl}(\theta) := \frac{1}{n} \sum_{j=1}^n l_{icl}(\theta; Z_j)$$

$$L_{icl}(\theta) = \mathbb{E}_{\substack{P \sim \pi \\ Z \sim P^{\otimes N+1}}} [l_{icl}(\theta; Z)]$$

Parameter space:

$$\Theta_{L,M,D',B} := \left\{ \theta = (\theta_{\text{actn}}^{(1:L)}, \theta_{\text{mlp}}^{(1:L)}) : \max_{l \in [L]} M^{(l)} \leq M, \max_{l \in [L]} D^{(l)} \in D', \|\theta\| \leq B \right\}$$

$$\text{where } \|\theta\| := \max_{l \in [L]} \left\{ \max_{m \in [M]} \{ \|Q_m^{(l)}\|_{\text{op}}, \|K_m^{(l)}\|_{\text{op}} \} + \sum_{m=1}^M \|V_m^{(l)}\|_{\text{op}} + \|W_1^{(l)}\|_{\text{op}} + \|W_2^{(l)}\|_{\text{op}} \right\}.$$

Thm 20 (Generalization for Pretraining): With prob at least $1-\xi$, the ERM solution $\hat{\theta}$ satisfies

$$L_{\text{cl}}(\hat{\theta}) \leq \inf_{\theta \in \Theta_{L,M,D',B}} L_{\text{cl}}(\theta) + \mathcal{O}\left(B_y^2 \sqrt{\frac{L^2(MD^2 + DD')\alpha + \log(1/\xi)}{n}}\right)$$

↖ number of prompts

bound on θ bound on H

$$\text{where } \alpha = \log(2 + \max\{B, R, B_y\}).$$

Proven using a chaining argument.

Thm 21 (Pre-Training Transformers for In-Context Linear Regression): Suppose $P \sim \pi$ is a.s. well-posed for linear regression. Then for $N \geq \tilde{O}(d)$, with prob. at least $1-\epsilon$, the solution $\hat{\theta}$ of ERM with $L = O(k \log(kN/\sigma))$ layers, $M=3$ heads, $D'=0$ (attention-only), and $B = O(\sqrt{kd})$ satisfies

$$L_{\text{ICL}}(\hat{\theta}) - \mathbb{E}_{P \sim \pi} \mathbb{E}_{(x,y) \sim P} \left[\frac{1}{2} (y - \underbrace{\langle w_p^*, x \rangle}_{\substack{\uparrow \\ \text{best linear} \\ \text{predictor}}})^2 \right] \leq \tilde{O} \left(\sqrt{\frac{k d^2 + \log(1/\epsilon)}{n}} + \frac{d \sigma^2}{N} \right)$$

where $\tilde{O}$ hides polylog factors in $k, N, 1/\sigma$.

- Remarks

- They showed that Transformers can implement GD by proposing values of the Q , K , V , and W matrices, but will a trained Transformer find this?
- This paper constructs an approximation to GD step with Transformers, but (Alyjarek et al., 2023; von Oswald et al., 2023) can exactly replicate the GD step